Adiabatic process:

An adiabatic process is defined as one in which there is no exchange of heat (q) between the system and surrounding during the process. Those processes in which no heat can flow into or out of the system are called ***adiabatic processes.*** This condition is attained by thermally insulating the system. In an adiabatic process if work is done by the system its temperature decreases, if work is done on the system its temperature increases, because, the system cannot exchange heat with its surroundings.

For an adiabatic process q = 0

Isothermal process :

An isothermal process is defined as one in which the temperature of the system remains constant, during the change from its initial to final state. The system exchanges heat with its surrounding and the temperature of the system remains constant. For this purpose the experiment is often performed in a thermostat.

For an isothermal process $dT = 0$

Isobaric process

An isobaric process is defined as one in which the pressure of the system remains constant during its change from the initial to final state.

For an isobaric process $dP = 0$.

Isochoric process

An isochoric process is defined as the one in which the volume of system remains constant during its change from initial to final state. Combustion of a fuel in a bomb calorimeter is an example of an isochoric process.

For an isochoric process, $dV = 0$.

Cyclic process:

When a system returns to its original state after completing a series of changes, then it is said that a cycle is completed. This process is known as a cyclic process.

For a cyclic process dU = 0, dH = 0, dP=0, dV=0, dT=0

Table: 7.2 Overview of the process and its condition

Process	Condition
Adiabatic	q = 0
Isothermal	dT= 0
Isobaric	dP= 0
Isochoric	dV= 0
Cyclic	dE= 0, dH = 0, dP=0, dV=0 , dT=0

State functions, path functions:

State function

A thermodynamic system can be defined by using the variables P, V, T and `n'. A state function is a thermodynamic property of a system, which has a specific value for a given state and does not depend on the path (or manner) by which the particular state is reached.

11th Std Chemistry 186-228.indd 190 06-01-2021 15:46:11